

SECTION 3
ACMV INSTALLATION

GENERAL SPECIFICATION

DUCTWORK AND FITTINGS

Material: nozzle and housing constructed of aluminium and swivel mechanism of GI steel.

A Twist element is incorporated at the inlet of the swivel jet nozzle element is incorporated at the inlet of the swivel jet nozzle.

The twist nozzles shall have performance data on air throw distance, sound power level and pressure loss against air flow rate.

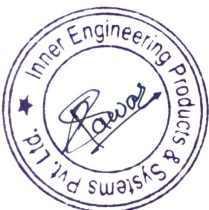
Material: nozzle and housing constructed of aluminium and swivel mechanism of GI steel, except the twist element is, made of polycarbonate material.

Test report must be submitted to guarantee the performance.

3.11 THERMAL AND ACOUSTIC INSULATION

3.11.1 General

- a. This section of the specification covers the supply and installation of the thermal and acoustic insulation required for ductwork.
- b. Preparation of the equipment and application of all insulating materials shall be carried out only by skilled tradesmen, experienced and competent in this type of work.
- c. In general, insulating materials shall be of best quality in their respective types, and shall have the following properties:
 - i) High thermal efficiency and low moisture absorption combined with adequate strength to prevent deterioration under normal operating conditions.
 - ii) Resistance to fibre and attack of algae, bacteria, Vermin and the growth of mould and fungi.
 - iii) Free from objectionable smells when dry or wet and shall not cause or promote corrosion or deterioration when in contact with galvanised metal, steel, iron, copper, bronze, brass, aluminium or PVC.
- d. All insulating materials shall be delivered to the site in a dry condition and shall be housed in a dry store until drawn up for use.
- e. Prior to the application of insulation, all surfaces shall be thoroughly cleaned of scale, rust, grease, oil and dirt. The application of the insulating materials to any part of the installation shall not be commenced until all pressure tests on the relevant part have been completed and approved.



DUCTWORK AND FITTINGS

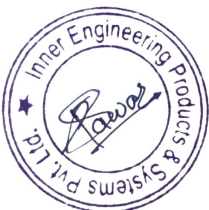
- f. All insulation materials, regardless of the methods of fixing, shall be in close contact with the surfaces to which they are applied. Edges and ends of insulating sections shall be arranged to butt up close to one another over the whole insulating surface. Joints of insulation shall be properly sealed. Pressure sensitive tapes shall not be accepted as joints having mechanical strength.
- g. Allowance for thermal movement shall be provided for in the application of insulation.

3.11.2

Thermal Insulation

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- a. Thermal insulation shall be of rigid fibreglass of 48kg/m^3 with thermal conductivity of not greater than 0.038 W/mK at 20°C mean temperature.
- b. The insulation shall be wrapped with fibre reinforced double sided aluminium foil vapour barrier. The fibre reinforcing grid shall not be greater than $10\text{mm} \times 10\text{mm}$. All joints in the aluminium foil shall be vapour sealed.
- c. Ductwork shall be insulated from hangers and supports generally as indicated in DW/144 : latest revision (HVAC).
- d. Fibreglass shall be adhered to duct with fire resistant and non water based adhesive. Adhesive shall be applied so that insulation conforms to duct surfaces uniformly and firmly.
- e. For ducts up to 450mm wide, the insulation shall be wrapped around the duct. For horizontal ducts with any side over 450mm , metal pins or nylon pins shall be stud welded using the capacitor discharge method without damaging the galvanising on the bottom surface at 400mm maximum centres. Then insulation shall be wrapped around. For vertical ducts with any side over 450mm , the pins shall be located on all sides of the duct at approximately 300mm centres. Protruding ends of pins shall be cut off flush after clips have been applied. The vapour barrier facing shall be thoroughly sealed with a vapour-barrier mastic or tape where the pins have pierced through.
- f. All joints shall be sealed with at least 75mm wide vapour barrier tape to provide a continuous external vapour seal. Prior to the application of tape all contact surfaces of vapour barrier foil shall be wiped clean of dust and grease using cloth and suitable solvent all in accordance with the manufacturer's recommendation.



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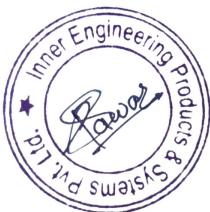
- g. All ductwork insulation shall extend over external flanges and stiffening.
- h. Insulation shall be applied so as to form a continuous heat barrier without gaps, cavities and openings. Care shall be exercised to ensure that the minimum thickness specified is maintained at corners, protrusions, etc.
- i. At the point of support, specially prepared blocks of hardwood or styrofoam material must be positioned to ensure integrity of vapour barrier by bonding the supports to the insulation.
- j. All externally insulated ductwork exposed in plant rooms, corridors and other areas subject to normal usage traffic shall be sheathed externally by means of 0.5 mm galvanised steel with all joints lapped and secured by self tapping screws at 100 mm centres.
- k. All insulation shall be applied so as to give a smooth, homogeneous and lineable surface. All rigid sections shall be concentric, and accurately matched for thickness. Steps and undulations in the surfaces shall not be acceptable. Any sections or slabs having damaged ends or edges shall be rejected.
- l. All insulation shall fit tight to surfaces to be covered, and all slabs and sections shall be built up close, butting edges being mitred, chamfered or shaped as necessary. Any minor interstices left in insulation shall be filled and sealed.
- m. The vapour barrier shall be sealed around all hangers or other projections through the insulation.
- n. Continuous insulation shall be provided through all sleeves and insulation joins shall be staggered with respect to joints on the associated ductwork system.
- o. Where ducts pass through walls and slabs, suitable fire stop system shall be provided for the duct including the fire damper. The continuity of the insulation shall be maintained throughout.

3.11.3

Acoustic Lining Ductwork

- a. All acoustic lining shall comply with Class 1 surface spread of flame and Class 0 rating when tested to BS 476.
- b. All acoustic lining shall have a thermal conductivity rating of not more than 0.036 W/m K at a mean temperature of 24°C and shall be of the semi rigid type manufactured from minimum 50mm thick

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mineral wool or fibre glass, having a density of not less than 48 kg/m³.

- c. Insulation shall be faced with perforated galvanised sheet. Insulation shall be adhered to the ducts with an approved type adhesive and fastened with welded pins and speed clips located at not more than 400mm centres.
- d. All joins between insulation butts shall have the surface layer lapped and adhered together or pop riveted as appropriate in the direction of the air flow.
- e. All end pieces shall be complete with sheet metal nosing and all corners shall be held firm with angles formed from galvanised sheet steel.
- f. Acoustic lining shall have sound absorption coefficients of not less than following:

Thickness (mm)	Frequency (hertz)				
	125	250	5000	1000	2000
25 (foil laminate)	.01	.19	.43	.71	.73
50 (foil laminate)	.22	.52	.73	.94	.94

- g. Acoustic lining in masonry ducts shall be as for sheet metal ducts as previously specified except that pressed 1 mm thick galvanised steel "Z" sections shall be located at approximately 1000mm centres to support the full weight of the butts in a horizontal direction.
- h. The galvanised steel "Z" sections shall be firmly attached to the masonry wall with expanding type masonry anchors or at not more than 450mm centres.
- i. Pins and speed clips shall be used as for ductwork except that approved cement type fixings shall be used for direct and firm attachment of pins to the masonry wall in lieu of welded type.

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